The Three Cheap Fixes

MCO-1D instrumentation brief · what each one adds, and why it is worth doing
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The one defect all three fix

The hypothesis library scores **the aircraft's reaction**. It never scores **the wing**. Of 141 scored metrics, 119 read airframe state and 16 read motor telemetry at the root; three read a *commanded* angle rather than a measured one, one reads a sensor that is not fitted, and two read a quantity that has no source anywhere in 92,587 lines.

Why that is worse than it sounds

When a hypothesis fails, the library cannot tell you which of these two happened:

(a) the wing did exactly what was commanded, and the aerodynamic reasoning behind the hypothesis is wrong;
(b) the aerodynamic reasoning is fine, and the wing never did what was commanded.

Both produce the same telemetry. Every negative result arrives with two explanations and no instrument to choose between them — so a failed hypothesis teaches almost nothing, and a *passed* one may have passed for the wrong reason. **The three fixes below exist to separate (a) from (b).** That is their real product; the individual hypotheses they unblock are a bonus.

Summary

	Fix	Adds	Cost	Added mass
1	**Wing pitch encoders, 8×**	Actual wing pitch angle, continuously, through the stroke	~$24–70	~16 g
2	**Servo-rail current shunt**	Measured servo power instead of a modelled number	~$8–15	~5 g
3	**TC-02 wired into scoring**	Stand-measured thrust as a baseline every thrust hypothesis is checked against	**$0**	**0 g**

Total: roughly **$32–85 and 21 g** — about half of one percent of the 4.189 kg all-up weight, against a library of 177 hypotheses that currently cannot see the wing at all.

FIX 1 Wing pitch encoders — 8 ×

Highest value in the build
~$24–70 · ~16 g total

What it measures

The actual angular position of each wing's pitch output, sampled continuously through the flap stroke. One sensor per wing, because with eight independently driven wings there is no such thing as a representative wing.

Why this particular quantity

`aoa_achieved_deg` is read twice in the library and written **never**. H-SERVO-1's entire lever is passive aeroelastic twist, computed as commanded AoA minus achieved AoA. The subtrahend does not exist, so the hypothesis cannot score at all — and it gates H-INT-1 and H-SUB-1 behind it.

There is a second, blunter reason. At line 11696, inside the in-flight full-range-of-motion check:

```
achieved[target_deg] = target_deg   # commanded = achieved (no encoder)
```

The routine that verifies each servo reached its commanded angle verifies it **by assuming it did**. That check currently cannot fail, which means it is not a check. An encoder makes it one.

What it unblocks

Effect	Detail
H-SERVO-1 becomes scorable	Commanded minus achieved *is* the passive-twist measurement. Direct, no inference.
H-INT-1 and H-SUB-1 unblock	Both gate on H-SERVO-1. H-INT-1's modelled benefit currently spans **0.1% to 32%** — a range wide enough to be no prediction at all.
The two-explanations problem closes	Not just for these three, but for **all 141 metrics**. Any hypothesis that fails can now be sorted into 'physics wrong' or 'wing did not comply'. This is the largest single gain and it is not specific to any hypothesis.
The wear modality gets a direct signal	Hinge slop and bearing wear show up as commanded-minus-achieved drift at *zero aerodynamic load* — measurable on the ground, trending over the airframe's life. Far cleaner than the back-EMF route, which currently has a suspected unit error.
Servo rate becomes measured	The 244°/s-vs-407°/s shortfall is presently a datasheet figure against a demand calculation. An encoder gives the real loaded slew rate on the real mechanism.

Part options

Option	How it works	Trade
AS5600L *recommended*	Magnetic rotary encoder, 12-bit absolute (4096 steps). A diametrically magnetised magnet on the shaft end, sensor on a fixed bracket 0.5–3 mm away, on-axis. Contactless.	**Programmable I^2C address** — all eight sit on one bus with no multiplexer. This is the variant to order; the plain AS5600 has a fixed address of 0x36 and eight of them will collide.
AS5600-ASOM	Same sensor, analog output mode, read through an external ADC.	Needs 2 × ADS1015 (4 channels each, ~3300 sps). Works, but it is more parts and more wiring than the AS5600L route for no gain.
AS5048A	14-bit, SPI, daisy-chainable — all eight read in one ~20 µs transaction.	Cleanest electrically and the fastest, but roughly 3–4× the unit cost. Worth it only if I^2C bandwidth turns out to be tight.

Option	How it works	Trade
Ratiometric Hall (SS495A / A1324)	Analog voltage against field angle, with a diametric magnet.	Cheapest and lightest, but nonlinear, temperature-sensitive, and only good over a limited arc. Adequate for a feasibility bench test, not for flight data you intend to score hypotheses against.

Sampling requirement

Flap frequency is 1.87 Hz. The twist waveform is not a sinusoid — the pitch reversal at each stroke end is the fast part, and it is the part that matters, because that is where passive twist either helps or does not. Budget harmonic content to roughly the eighth harmonic (≈15 Hz), which puts Nyquist at 30 Hz, and then oversample: **200 Hz per channel** is a sound target and well within what eight AS5600L devices on one bus will deliver.

Mounting and mass

- Magnet (6 × 2.5 mm diametric N35) bonded to the end of the pitch output shaft, on-axis. ~0.5 g.
- Sensor on a small fixed bracket, air gap 0.5–3 mm. No contact, no added friction, nothing in the load path. ~0.6 g with the breakout trimmed, plus wire.
- **Under 2 g per wing, ~16 g airframe-wide** — 0.38% of all-up weight. Compare the pre-torsion spring, which was dismissed partly because it cost 120 g.

> **One real risk — test it before committing eight installs**
>
> Each wing's pitch axis sits near a brushless motor. A magnetic rotary encoder reads a magnetic field, and a motor stator produces one.
>
> **Bench-test a single sensor beside a running motor at full current before ordering or installing eight.** If there is corruption, the mitigations are siting the magnet further from the stator, a mu-metal shield, or falling back to the optical or SPI route. This is a one-afternoon test that de-risks the whole fix, and it should happen first.

FIX 2 Servo-rail current shunt

Single install
~$8–15 · ~5 g

What it measures

Instantaneous current drawn by the 7.4 V servo rail feeding all sixteen AGFRC A27CHR units — and therefore servo power, continuously.

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Why it matters more than one sensor usually would

Servo power is the **denominator of every endurance hypothesis in the library**, and it is currently a modelled number rather than a measured one. It enters `int_optimal_duty()` directly:

```
p_aero  = max(1.0, p_cruise_w - p_servo_w - p_avionics_w)
p_glide = INT_GLIDE_SERVO_FRAC * p_servo_w + p_avionics_w
```

`p_servo_w` appears in both the aerodynamic power remainder *and* the glide-phase power, with opposite sign effects. That is why H-INT-1's predicted benefit spans **0.1% to 32%**: the uncertainty in one modelled input propagates to a two-and-a-half-order-of-magnitude spread in the output. **An $8 part collapses that range to a number.**

Part options

Option	How it works	Trade
INA226 + external shunt *recommended*	16-bit I^2C current/voltage/power monitor across a low-value shunt. Conversion times down to 140 µs, with programmable averaging.	Accurate, cheap, and fast enough (~3.5 kHz raw) to see within-stroke structure, not just a mean. Size the shunt around 1 mΩ for headroom over the sixteen-servo peak.
ACS758	Hall-effect current sensor. Galvanically isolated, no shunt burden voltage.	Simpler to wire and immune to common-mode issues, but noisier and more temperature-drifty. Fine for a mean, poor for the within-stroke detail that is the interesting part.

What it unblocks

- **The other half of H-SERVO-1.** Fix 1 gives the twist angle; Fix 2 gives the power that twist was supposed to save. Neither is sufficient alone — the hypothesis is a claim about the relationship between them.
- **H-DUTY-1 verification.** Stroke duty-cycle shaping is a servo-power argument end to end.
- **Regenerative accounting on the servo side.** H-REGEN-2 currently counts what comes back on the motor bus; it does not count what the servo rail spends during the same event.

> **Fixes 1 and 2 are worth more together than apart**
>
> Sample both at 200 Hz and you can overlay the measured twist angle on the measured servo current within a single stroke.
>
> If passive aeroelastic twist is doing what H-SERVO-1 claims, there will be a **visible notch in servo current at the point in the stroke where the wing twists under aerodynamic load rather than against the servo**. That is a direct observation of the mechanism, not an inference from endurance. It is the difference between knowing the aircraft flew longer and knowing why.

FIX 3 TC-02 wired into hypothesis scoring

Software only
$0 · 0 g

What already exists

TC-02 is a static thrust test on an HX711 load cell, and it is **real, finished code** — 138 references through the library, live force streaming, cycle-averaging against flap frequency, and a CSV that already records:

```
freq_hz, amp_deg, cascade_phase_deg, profile, mean_force_n, lift_n, load_cell_n, peak_pos_n,
peak_neg_n, peak_to_mean, CT, advance_J, motor_I_A, n_cycles, hover_margin_n
```

That is the only genuinely instrumented measurement of the wings anywhere in the project.

What is missing

The CSV is written and **never read back**. Prerequisite plumbing across the entire library totals four `prereq` entries, one `gated_by`, two `requires_h` and one `depends_on` — **eight links for 137 hypotheses**. The stand rig and the hypothesis library are two systems that do not speak to each other.

What to build

Piece	What it does
a. Loader	Reads the TC-02 CSV into the hypothesis evidence store, keyed on (freq, amplitude, cascade phase, profile).
b. Prerequisite links	Every thrust-dependent hypothesis gates on a TC-02 baseline existing for its operating point. No baseline, no score — rather than scoring against a modelled thrust and calling it evidence.

Piece	What it does
c. Comparator	Flags when in-flight behaviour departs from the stand prediction beyond a threshold. A departure is information either way: it is either an unmodelled forward-flight effect or a stand artefact, and both are worth knowing.

Why this is the highest leverage per hour spent

It tests hypotheses **before an airframe is at risk**. Nothing has flown; H-PERF-6 has not run; `HOVER_CAPABLE` is still `None`. The stand is available now.

> **The specific result the stand can settle this week**
>
> H-BUS-1 predicts that inertial power ripple goes to **exactly zero at cascade phase 22.5° and 45°** for eight wings, and is at its *worst* at 180° — the opposite of the intuitive 'split into halves' answer.
>
> `cascade_phase_deg` is already a TC-02 sweep axis, and `motor_I_A` is already a recorded column. The prediction is a **current-ripple null** at two specific phase angles, visible on a bench rig that exists, with no flying and no risk.
>
> If the null is there, the electrical spring is real and the mechanical spring stays deleted. If it is not, that needs to be known before the airframe is built around the assumption.

Honest limit

Static thrust is not forward flight. The stand validates the thrust map, the phase-cancellation prediction, and the peak-to-mean force structure. It cannot validate anything that depends on forward airspeed — which is most of the library, including the entire intermittency family and all four forward-flight SAR hypotheses. **It is a floor under the library, not a substitute for flying.**

Order of work

	Do this	Why in this position
1	**Wire TC-02 into scoring** (Fix 3)	No hardware, no cost, no mass, and it can be done now while the airframe is still being built. It also gives the other two fixes somewhere to report into.
2	**Bench-test one magnetic encoder beside a running motor**	De-risks Fix 1 for the cost of one part and one afternoon. Do not order eight until this passes.
3	**Fit the servo-rail shunt** (Fix 2)	One install, one part, immediate payoff. Independent of everything else — nothing blocks it.
4	**Fit the eight encoders** (Fix 1)	Largest gain, but gated on step 2 and the most work. Worth doing before first flight if the schedule allows, because H-PERF-6 is far more informative with them than without.

What these three do *not* fix

Stated plainly so the fixes are not oversold.

Still open	Why these fixes do not touch it
No aerodynamic measurement of any kind	There is still no pressure tap, no tuft, no flow-state sensing. Separation, reattachment and bubble behaviour remain invisible. **H-LSB-1 stays untestable in flight** regardless of all three fixes.
$C_m = 0.25$ is still modelled	That is an XFOIL run on the exported modified-S1223 coordinates — a separate item, and still the cheapest open thing in the project. It sets the servo load, the pivot choice and V_{max}.
The 4.0 g structural limit is still an estimate	That is TC-05, static load testing. Every dive number scales with it, including the recovery-window closure speed.
The servo rate shortfall stays open	244°/s loaded against 407°/s demanded at hover is not an instrumentation problem. Measuring it more precisely does not close it. Either the hover demand comes down or the architecture changes.
Forward-flight hypotheses still need flight	The stand cannot reach them, and encoders on the ground cannot either. H-PERF-6 remains the event that starts converting the library.

What you get for ~$32–85 and 21 g

Three hypotheses move from unscorable to scorable. One prediction (the H-BUS-1 phase null) becomes testable on a bench this week with no hardware purchase at all.

And every one of the 141 scored metrics gains the ability to distinguish a wrong idea from a wing that did not comply — which is the difference between a library that learns from failure and one that only accumulates it.

Part numbers and the AS5600L programmable-address behaviour should be confirmed against current datasheets before ordering; prices are approximate and were not quoted from a supplier.